

PROJECT REPORT

Data Fundamentals with Data Warehousing, SQL and Python

B.Tech Program in ECSc

Submitted By

Name:-Mobashshar Faiz

Roll No: 2330379



**Kalinga Institute of Industrial Technology
(Deemed to be University)
Bhubaneswar, India**

Spring 2024-2025

Title : Hospital Patient Data Analysis

I) **Problem Statement** : Hospitals generate a vast amount of patient data on a daily basis, including details such as age, gender, diagnosed diseases, assigned doctors, and treatment costs. However, this data often remains underutilized due to the lack of structured analysis and visualization. Without proper insights, it becomes difficult for hospital management to identify prevalent diseases, efficiently distribute workload among doctors, and monitor treatment cost patterns.

The absence of data-driven decision-making can lead to inefficiencies such as uneven doctor workload, delayed identification of common health issues, and suboptimal allocation of resources. Therefore, there is a need for a system that can systematically analyze patient data to extract meaningful insights.

This project aims to address these challenges by analyzing hospital patient data using SQL, Excel, and Python. The objective is to identify disease trends, evaluate doctor workload, and understand treatment cost distribution through structured queries and visualizations. The insights derived from this analysis can assist hospital management in improving operational efficiency, enhancing patient care, and making informed strategic decisions.

II) Proposed Solutions/ Features :

The proposed solution focuses on analyzing hospital patient data using a combination of SQL, Excel, and Python to extract meaningful insights and present them in an easy-to-understand format. The system is designed to process structured patient data and provide actionable information to support hospital management decisions.

Key Features

1. Disease Pattern Analysis

The system identifies the frequency of different diseases among patients using SQL queries and Python analysis. This helps in understanding which diseases are most common and require more attention and resources.

2. Doctor Workload Analysis

By grouping patients based on assigned doctors, the system evaluates the workload distribution among doctors. This helps in identifying overburdened doctors and enables better resource allocation.

3. Treatment Cost Analysis

The project calculates statistical measures such as average, minimum, and maximum treatment costs. This helps in understanding cost patterns and identifying expensive treatments.

4. Data Visualization

The system uses Python libraries to generate visual representations:

- Pie chart for disease distribution
- Bar chart for doctor workload

These visualizations make it easier to interpret data quickly.

5. Excel-Based Summary (Pivot Table)

An Excel pivot table is used to summarize disease counts, providing a quick and interactive way to explore data.

6. Multi-Tool Integration

The project integrates:

- SQL for querying
 - Excel for summarization
 - Python for analysis and visualization
- This demonstrates a complete data analysis workflow.

III) Screenshots:

- SQL Outputs :

patient_id	age	gender	disease	doctor	treatment_cost
1	45	Male	Diabetes	Dr Shah	5000
2	30	Female	Fever	Dr Patil	2000
3	55	Male	Heart Disease	Dr Shah	20000
4	22	Female	Flu	Dr Kulkarni	1500
5	40	Male	Diabetes	Dr Shah	4500
6	60	Female	Heart Disease	Dr Patil	22000
7	35	Male	Fever	Dr Kulkarni	1800
8	28	Female	Flu	Dr Shah	1600

disease	patient_count
Diabetes	3
Fever	3
Heart Disease	2
Flu	2

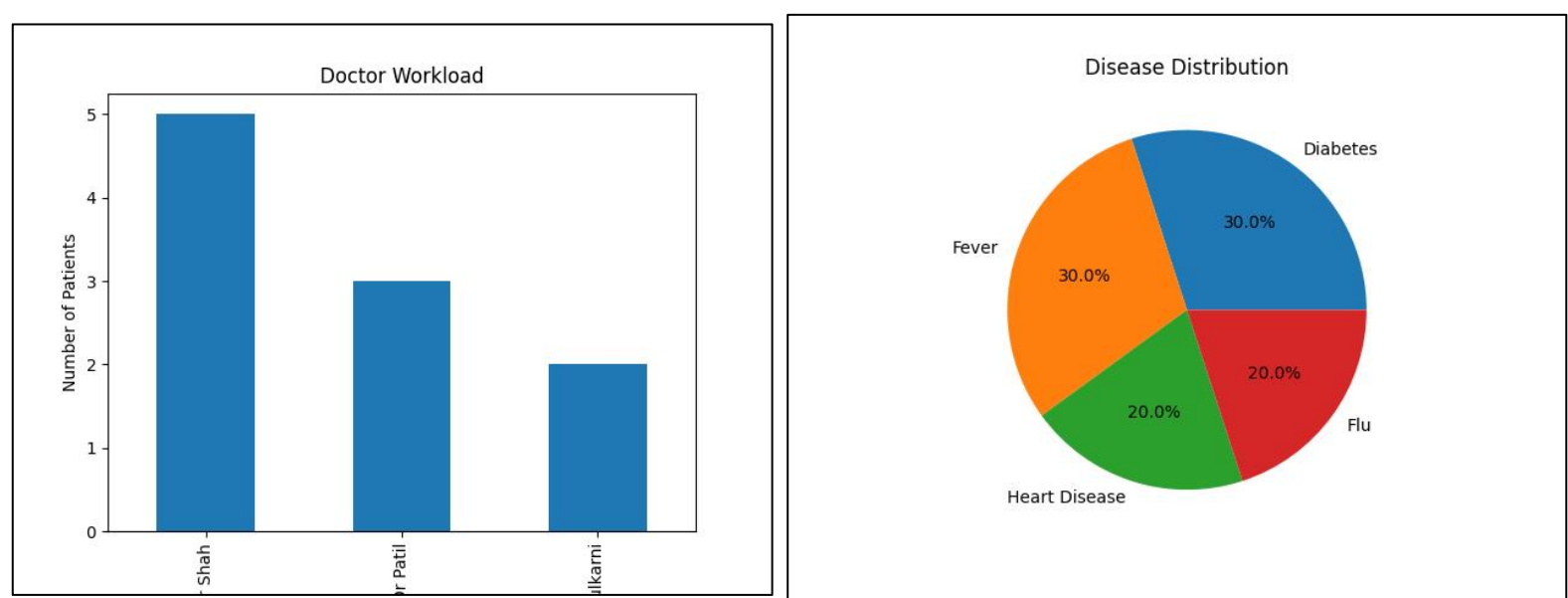
doctor	total_patients
Dr Shah	5

avg_cost
6570.0000

- Excel Outputs :

disease	Count of patient_id
Diabetes	3
Fever	3
Flu	2
Heart Disease	2
Grand Total	10

- Python Outputs :



IV) Tech Stack :

The project utilizes a combination of tools and technologies to perform data analysis, querying, and visualization effectively. Each component of the tech stack plays a specific role in extracting and presenting insights from the hospital dataset.

1. Python

Python is used as the primary programming language for data analysis and visualization. It provides powerful libraries that simplify data processing and graphical representation.

- **Pandas:** Used for data loading, cleaning, and analysis
- **Matplotlib:** Used for creating visualizations such as pie charts and bar graphs

2. SQL (MySQL)

SQL is used for structured data querying and analysis. The project uses MySQL Workbench to execute queries and extract insights from the dataset.

- Data storage using relational tables
- Aggregation using GROUP BY
- Statistical queries like COUNT() and AVG()
- Identifying trends such as most common diseases and doctor workload

3. Microsoft Excel

Microsoft Excel is used for quick data summarization and interactive analysis.

- Creation of Pivot Tables
- Disease-wise patient count analysis
- Easy visualization of summarized data

4. CSV (Comma-Separated Values)

The dataset is stored in CSV format, which serves as a lightweight and portable data source.

- Easy to import into Python, SQL, and Excel
- Structured tabular format for consistency

V) Unique Points :

This project stands out due to its practical approach and integration of multiple tools to perform end-to-end data analysis. Unlike basic projects that rely on a single technology, this solution combines SQL, Excel, and Python to demonstrate a complete data analytics workflow.

Key Unique Aspects

1. Multi-Tool Integration

The project effectively integrates SQL for querying, Excel for summarization, and Python for analysis and visualization, showcasing versatility in handling data across platforms.

2. Real-World Use Case

The dataset and analysis are based on a real-world healthcare scenario, making the project relevant and applicable to hospital management systems.

3. Insight-Driven Approach

Instead of just displaying data, the project focuses on extracting meaningful insights such as disease trends, doctor workload distribution, and cost patterns.

4. Visual Representation of Data

The use of charts (pie and bar graphs) makes complex data easy to understand, enabling better decision-making.

5. Structured and Scalable Design

The project is organized in a modular way (CSV → SQL → Excel → Python), making it easy to scale or extend for larger datasets and real-time systems.

VI) Future Improvements :

1. Machine Learning Integration

Implement predictive models to forecast diseases based on patient history and trends, enabling proactive healthcare decisions.

2. Interactive Dashboard

Develop a web-based dashboard using tools like Streamlit or Power BI for real-time data visualization and user interaction.

3. Real-Time Data Integration

Connect the system to live hospital databases to analyze data dynamically instead of using static CSV files.

4. Patient Segmentation Analysis

Extend the project to analyze patients based on age groups, gender, or medical history for more detailed insights.

5. Automated Reporting System

Generate automated reports for hospital administrators, reducing manual effort and improving efficiency.